

Know.	App.	Think.	Comm.
/23	/17	/5	/3

Knowledge and Understanding [23 marks]

1. **Multiple Choice:** Write the CAPITAL LETTER corresponding

to the correct answer on the line provided. [1 mark each]

i. The average rate of change of $f(x) = x^2 + 3x + 4$ over the interval $0 \leq x \leq a$ is:

- (A) $3a$ (B) $a^2 + 3$ (C) $a + 3$ (D) none of these

C

ii. The function $f(x)$ has a jump discontinuity at $x=a$ if

(A) $\lim_{x \rightarrow a} f(x)$ exists (B) $f(a)$ is undefined

(C) $\lim_{x \rightarrow a} f(x) = f(a)$ (D) $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$

D

iii. Evaluate the limit, if it exists: $\lim_{x \rightarrow \infty} \frac{(3x^2 - 1)^2}{2x^4 - 3x^2 + 1}$

- (A) $\frac{3}{2}$ (B) ∞ (C) 0 (D) $\frac{9}{2}$

D

iv. If $f(x) = x^2 - 1$, the value of $\left. \frac{dy}{dx} \right|_{x=2}$ is

- (A) 2 (B) 4 (C) 0 (D) 1

B

2. Given the function $y = f(x)$, determine the following: [1 mark each]

~~a)~~ $\lim_{x \rightarrow 5} f(x) = \text{omit}$

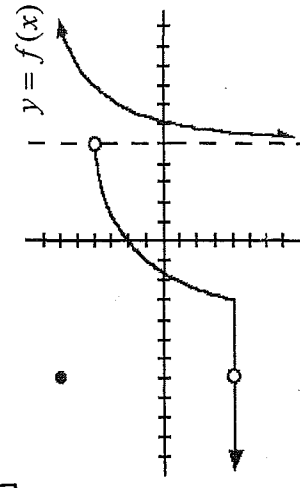
b) $\lim_{x \rightarrow 5^+} f(x) = -\infty$

c) $\lim_{x \rightarrow 5} f(x) = 0 \text{ N.E.}$

d) $f(5) = \text{undefined}$

e) $f(-7) = 6$

f) $\lim_{x \rightarrow -7} f(x) = -4$



3. Evaluate the following limits. [10 marks]

a) $\lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1}$ ✓✓

$$= \lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{\sqrt{x}(x-1)} = \lim_{x \rightarrow 1} \frac{(1-\sqrt{x})(1+\sqrt{x})}{\sqrt{x}(x-1)(1+\sqrt{x})} = \lim_{x \rightarrow 1} \frac{1-x}{\sqrt{x}(x-1)(1+\sqrt{x})} = \frac{-1}{2}$$

b) $\lim_{x \rightarrow 1} \frac{x^3 - 5x^2 - x + 5}{x^3 + 2x^2 - x - 2}$ ✓✓

$$= \lim_{x \rightarrow 1} \frac{x^3 - x - 5x^2 + 5}{x^3 - x + 2x^2 - 2} = \lim_{x \rightarrow 1} \frac{x(x^2-1) - 5(x^2-1)}{x(x^2-1) + 2(x^2-1)} = \lim_{x \rightarrow 1} \frac{(x^2-1)(x-5)}{(x^2-1)(x+2)} = \frac{-4}{3}$$

c) $\lim_{x \rightarrow 1} \frac{\sqrt[4]{x} - 1}{\sqrt[8]{x} - 1}$ ✓✓✓

Let $t = \sqrt[8]{x}$ $t \rightarrow 1$
 $t^2 = \sqrt[4]{x}$
 $\lim_{t \rightarrow 1} \frac{t^2 - 1}{t - 1} = \lim_{t \rightarrow 1} \frac{(t-1)(t+1)}{t-1} = 2$

4. Sketch a single function $y = f(x)$ that has all of the following properties: ✓✓✓

i. $\lim_{x \rightarrow 2^+} f(x) = -\infty$

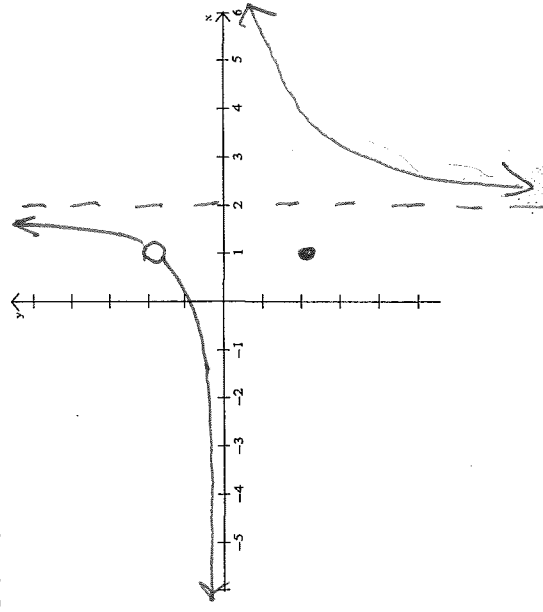
ii. $\lim_{x \rightarrow 2^-} f(x) = \infty$

iii. $\lim_{x \rightarrow -\infty} f(x) = 0$

iv. $\lim_{x \rightarrow 1^+} f(x) = 2$

v. $\lim_{x \rightarrow 1^-} f(x) = 2$

vi. $f(1) = -2$



Application [17 marks]

1. The position of a particle moving along the x-axis is given by $s(t) = t^2 - 5t + 4$ where t is the elapsed time in seconds.

- (a) Calculate the average velocity of the particle from $t = 2s$ to $t = 5s$. ✓✓

$$V_A = \frac{s(5) - s(2)}{5 - 2} = \frac{25 - 25 + 4 - (4 - 10 + 4)}{3} = 2$$

- (b) Calculate the instantaneous velocity of the particle when $t = 2s$. ✓✓✓

$$V_I = \lim_{h \rightarrow 0} \frac{(2+h)^2 - 5(2+h) + 4 - (4 - 10 + 4)}{h} = \lim_{h \rightarrow 0} \frac{4 + 4h + h^2 - 10 - 5h + 4 - 4 + 10 - 4}{h} = \lim_{h \rightarrow 0} \frac{h^2 - h}{h} = \lim_{h \rightarrow 0} h(h-1) = -1$$

2. A tangent to the parabola $y = \sqrt{ax+b}$ is perpendicular to $3x + y + 4 = 0$ at point (4,3). Determine the values of a and b . ✓✓✓✓

$$m_t = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \rightarrow y = -3x - 4 \Rightarrow m_t = \frac{1}{3}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{a(x+h)+b} - \sqrt{ax+b}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(\sqrt{a(x+h)+b} - \sqrt{ax+b})(\sqrt{a(x+h)+b} + \sqrt{ax+b})}{h(\sqrt{a(x+h)+b} + \sqrt{ax+b})}$$

$$= \lim_{h \rightarrow 0} \frac{a(x+h) + b - ax - b}{h(\sqrt{a(x+h)+b} + \sqrt{ax+b})}$$

$$= \frac{a}{2\sqrt{ax+b}} = \frac{1}{3} \Rightarrow 3a = 2\sqrt{ax+b}$$

$$3a = 2\sqrt{4a+b} \quad (2)$$

$$\text{Also, } 3 = \sqrt{4a+b}$$

$$9 = 4a + b$$

$$b = 9 - 4a \quad (1)$$

Sub (1) into (2)

$$3a = 2\sqrt{4a + 9 - 4a}$$

$$3a = 6$$

$$\boxed{a = 2}$$

$$\boxed{b = 1}$$

3. Function $g(x) = \frac{x-3}{\sqrt{5x+1} - \sqrt{3x+7}}$ is given. For what value of the constant k is the function

$$f(x) = \begin{cases} g(x) & x \in D_g \\ k & x = 3 \end{cases} \text{ continuous on its domain? } \checkmark \checkmark \checkmark$$

we need to evaluate $\lim_{x \rightarrow 3} g(x)$

$$\lim_{x \rightarrow 3} g(x)$$

$$= \lim_{x \rightarrow 3} \frac{x-3}{\sqrt{5x+1} - \sqrt{3x+7}}$$

$$= \lim_{x \rightarrow 3} \frac{(x-3)(\sqrt{5x+1} + \sqrt{3x+7})}{(\sqrt{5x+1} - \sqrt{3x+7})(\sqrt{5x+1} + \sqrt{3x+7})}$$

$$= \lim_{x \rightarrow 3} \frac{(x-3)(\sqrt{5x+1} + \sqrt{3x+7})}{5x+1 - 3x-7}$$

$$= \lim_{x \rightarrow 3} \frac{(x-3)(\sqrt{5x+1} + \sqrt{3x+7})}{2(x-3)}$$

$$= \frac{4+4}{2}$$

$$= 4$$

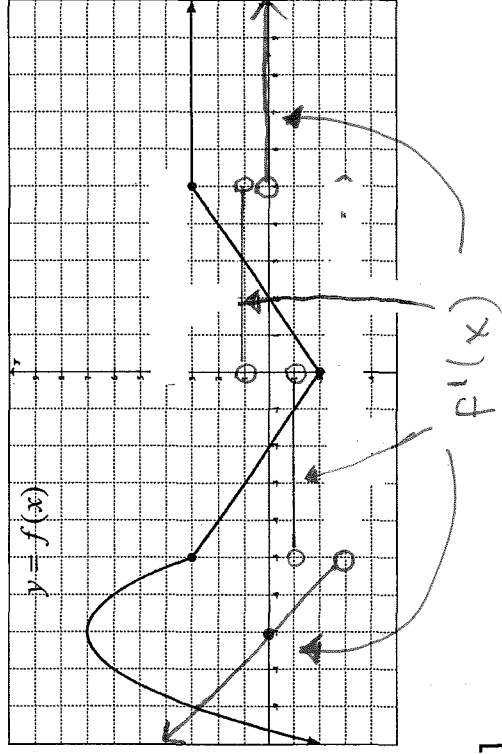
$\therefore k = 4$ for

$f(x)$ to be

continuous on

its domain

4. Graph the derivative function on the grid for the function $y=f(x)$ given below. ✓✓✓



Thinking [5 marks]

1. Given the function $f(x) = \frac{x^3 + ax^2 + bx + 5}{x^3 + 2x^2 - x - 2}$, determine whether there are numbers a and b such that both $\lim_{x \rightarrow 1} f(x)$ exists, and $\lim_{x \rightarrow -1} f(x)$ exists. If so, evaluate both limits. ✓✓✓✓

$$f(x) = \frac{x^3 + ax^2 + bx + 5}{x^3 + 2x^2 - x - 2} = \frac{x^2(x+2) - (x+2)}{x^3 + ax^2 + bx + 5} = \frac{(x+2)(x^2 - 1)}{(x+2)(x^2 - 1)}$$

$$= \frac{x^3 + ax^2 + bx + 5}{(x+2)(x^2 - 1)}$$

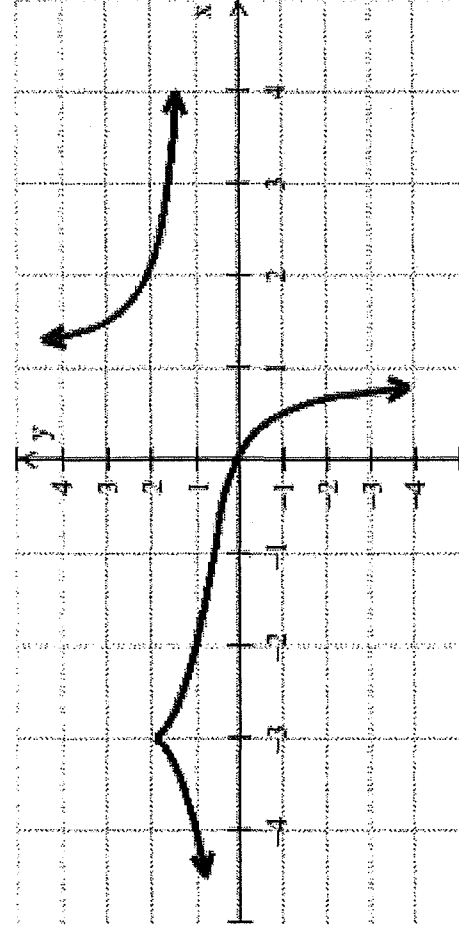
for $\lim_{x \rightarrow 1} f(x)$ and $\lim_{x \rightarrow -1} f(x)$

to exist $x+1$ and $x-1$ must be a factor of the numerator.

$$\lim_{x \rightarrow 1} f(x) = \frac{-4}{3} \quad \lim_{x \rightarrow -1} f(x) = -6$$

Communication [3 marks]

Determine at which value(s) of x the following function is not differentiable. Explain your reasoning clearly. [3 marks]



@ $x = -3$

* sharp corner. no

unique tangent line. * vertical asymptote $f(1)$ is not defined.

@ $x = 1$